

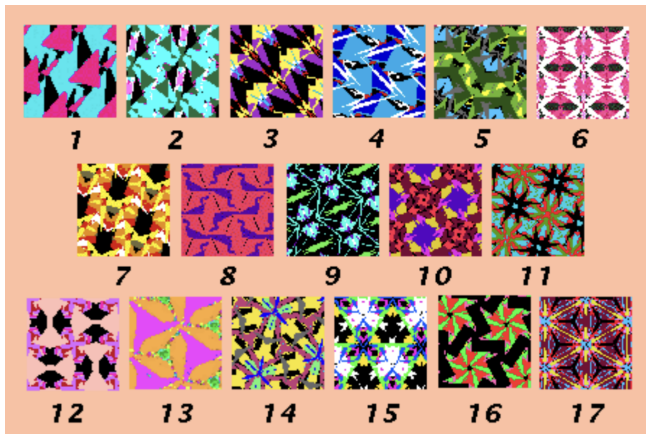
Group actions, algebra up to homotopy, and flavors of commutativity

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Groups and group actions



<https://www2.clarku.edu/faculty/djoyce/wallpaper/>

Poincaré Conjecture & Milnor's Question

Milnor's Questions

How many smooth structures are there on the n -sphere?

Theorem (Poincaré Conjecture: Smale–Freedman–Perelman)

If M is a homotopy n -sphere that is a manifold, then M is homeomorphic to S^n .

Kervaire–Milnor 1963

Definition

Let Θ_n be the group of diffeomorphism classes of n -spheres with addition connect sum.



ψ_n

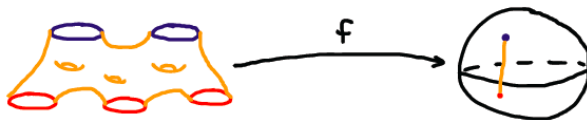
$$\Theta_n \xrightarrow{\psi_n} \pi_n^S / \text{Im}(J)$$

Pontryagin's Work

Definition

A framed n -manifold is an n -manifold with a continuous choice of basis for the normal vectors at every point

$$\{M^n \subset \mathbb{R}^{n+k}\} / \text{cobordism} \longleftrightarrow \pi_{n+k} S^k$$



$$\text{Framings on standard } S^k \longleftrightarrow \text{Im}(J)$$

Kervaire Invariant One Problem

Theorem (Kervaire–Milnor)

If $n \not\equiv 2 \pmod{4}$, then the map Ψ_n is surjective:

$$\Psi_n: \Theta_n \twoheadrightarrow \pi_n^S / \text{Im}(J).$$

If $n \equiv 2 \pmod{4}$, then there is an exact sequence

$$\Theta_n \xrightarrow{\Psi_n} \pi_n^S / \text{Im}(J) \xrightarrow{\Phi_n} \mathbb{Z}/2.$$

Theorem (H.–Hopkins–Ravenel,)

The map Φ_n is zero unless $n = 2, 6, 14, 30, 62$ or maybe 126.

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Theorem (H.–Hopkins–Ravenel, Lin–Wang–Xu)

*The map Φ_n is zero unless $n = 2, 6, 14, 30, 62$ **and** 126.*

Main Tool

Unexplored Structure in Equivariant Homotopy

Found new structure: equivariant commutative ring spectra have multiplication maps

$$\underbrace{R \otimes \cdots \otimes R}_n \rightarrow R,$$

but where the *group permutes the coordinates*.

What are these twisted products?

Main questions:

- 1 What sorts of structures do we see?
- 2 How do we understand it?

Arc of the talk

- 1 Describe commutativity and associativity as structure
- 2 Explore how twisted multiplications arise
- 3 Show how this new language has been applied

Poincaré's Fundamental Group

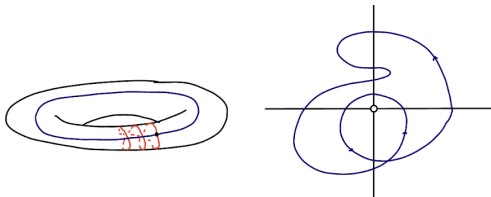
Loops

A loop on a pointed space X is a continuous function

$$\gamma: [0, 1] \rightarrow X$$

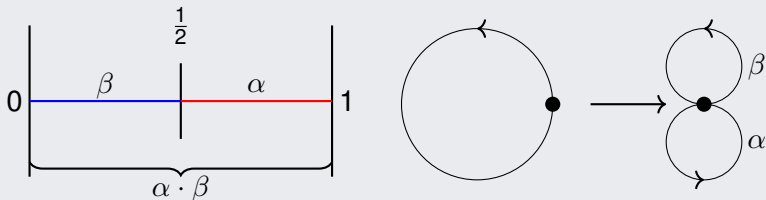
such that $\gamma(0) = \gamma(1) = x_0$.

A loop is a continuous pointed function $\gamma: S^1 \rightarrow X$.



Loop addition

Multiplication



Homotopy

A homotopy from γ_0 to γ_1 is a continuous function

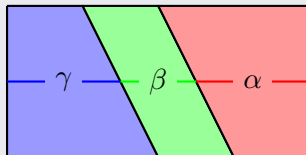
$$H: S^1 \times [0, 1] \rightarrow X$$

such that $H(z, 0) = \gamma_0(z)$, $H(z, 1) = \gamma_1(z)$, and $H(1, t) = x_0$.

Loop spaces

Loop spaces are “groups up to homotopy”

$$\alpha \cdot (\beta \cdot \gamma) \simeq (\alpha \cdot \beta) \cdot \gamma$$



Loops on X

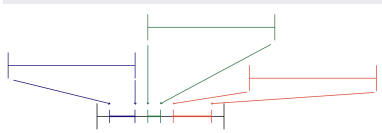
If X is a pointed space, then let $\Omega X = \text{Map}_*(S^1, X)$ be the space of continuous functions $S^1 \rightarrow X$ taking 1 to x_0 .

How is this a group?

What structure is present here?

Little intervals

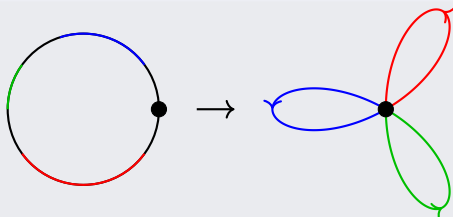
Embedding Intervals



Look at the space

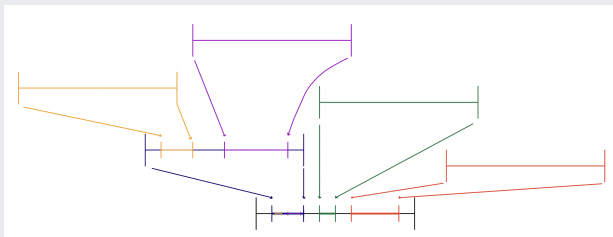
$$\mathcal{E}_1(n) = \text{Emb} \left(\coprod^n I, I \right).$$

Give new multiplications



Composition

Associativity



$$\mathcal{E}_1(3) \times (\mathcal{E}_1(2) \times \mathcal{E}_1(1) \times \mathcal{E}_1(1)) \rightarrow \mathcal{E}_1(4).$$

Permuting Coordinates

Have a Σ_n action on $\mathcal{E}_1(n)$

Little intervals operad

Definition (May)

An operad is a sequence of Σ_n -spaces $\mathcal{O}(n)$ together with composition maps

$$\mathcal{O}(n) \times \mathcal{O}(k_1) \times \cdots \times \mathcal{O}(k_n) \rightarrow \mathcal{O}(k_1 + \cdots + k_n)$$

that associate and that respect reordering the factors.

Definition

An $\mathcal{O}$ -algebra is a space X with maps $\mathcal{O}(n) \times X^{\times n} \rightarrow X$ compatible with composition.

Loop spaces as algebras

The $\mathcal{E}_k(n)$ assemble into an operad.

Loops are groups

The iterated loop space $\Omega^k X$ is an $\mathcal{E}_k(n)$ -algebra.

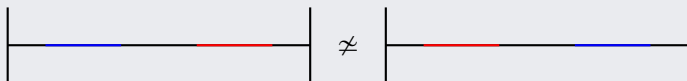
Theorem (May): Groups are loops

Group-like $\mathcal{E}_k(n)$ -algebras are iterated loop spaces.

Higher commutativity

Little intervals

The space $\mathcal{E}_1(2)$ is not path connected:



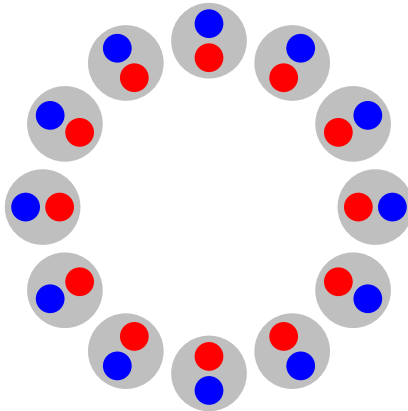
Little cubes

As k grows, the spaces

$$\mathcal{E}_k(n) = \text{Emb} \left(\prod_{i=1}^n I^k, I^k \right)$$

become more and more connected.

Higher loop spaces



Coherent Commutativity

$\mathcal{E}_\infty$

An $\mathcal{E}_\infty$ -operad has $\mathcal{E}_\infty(n)$ a contractible, Σ_n -free space.

What does this give us?

We have an essentially unique product $X^{\times n} \rightarrow X$ for all n .

Slogan

Ordinary $\mathcal{E}_\infty$ -operads structure “coherently commutative” ordinary multiplications, and there is only one, up to homotopy.

Equivariant Homotopy

Equivariant Homotopy

Homotopy theory for spaces with a G -action.

- For $H \subset G$, have “fixed points” X^H .
- If $K \subset H$, then have a map of fixed points $X^H \rightarrow X^K$

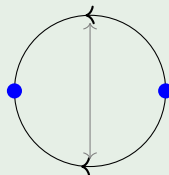
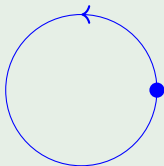
Shape for C_2

$$\begin{array}{c} X^{C_2} \rightarrow X \\ \quad \quad \quad \curvearrowright \\ \quad \quad \quad C_2 \end{array}$$

C_2 -equivariant 1-spheres

Example

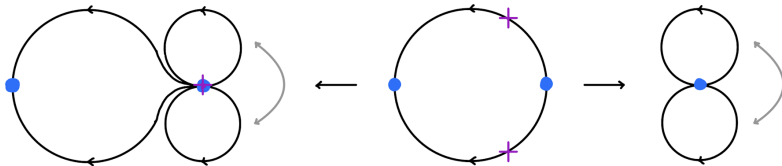
If $G = C_2$, then we have S^1 and S^σ , the unit complex numbers with complex conjugation.



Two kinds of loop spaces

Have $\Omega X = \text{Map}_*(S^1, X)$ and $\Omega^\sigma X = \text{Map}_*(S^\sigma, X)$.

Signed loop spaces



“Multiplications”

$$\Omega^\sigma X \times (\underbrace{\Omega X \times \Omega X}) \rightarrow \Omega^\sigma X \leftarrow (\underbrace{\Omega X \times \Omega X}).$$

No ordinary product

$\Omega^\sigma X$ does not have a multiplication! $\Omega^\sigma X \times \Omega^\sigma X \nrightarrow \Omega^\sigma X$

Structure on signed loops

Underlying

- Non-equivariantly ΩX is an ordinary loop-space
- C_2 acts as an anti-automorphism: the inverse

Fixed Points

- Have an action of ΩX on $(\Omega^\sigma X)^{C_2}$
- Have a “unit” map $\Omega X \rightarrow (\Omega^\sigma X)^{C_2}$.

Mackey Functors

The signed-loop action gives us a transfer map!

Twisted multiplications

General Framework

Have “twisted” multiplications

$$\text{Map}(\{1, \dots, n\}, X) \cong X^{\times n} \rightarrow X$$

where G acts on X and on the set $\{1, \dots, n\}$.

Definition

If $n = |G/H|$ for some subgroup H , then we can have a twisted multiplication map

$$\text{Map}(G/H, X) \rightarrow X.$$

This is a “norm from H to G ”.

Example: Galois Theory

Fundamental Theorem of Galois Theory

If $k \rightarrow K$ is a Galois field extension with Galois group G , then we have intermediate subfields K^H for any $H \subset G$ and relevant inclusions.

Transfers

If $J \subset H$, have a twisted sum, the trace / transfer:

$$\sum_{g \in H/J} g \cdot (-): K^J \rightarrow K^H.$$

Norms

If $J \subset H$, have a twisted product, the norm:

$$\prod_{g \in H/J} g \cdot (-): K^J \rightarrow K^H.$$

Equivariant commutativity

New kinds of multiplications

Equivariantly, we have the option for not only ordinary products but also twisted products.

Restating our goal

How do describe having coherently commutative norms?

N_∞ Operads

Definition (Blumberg–H.)

An N_∞ operad for G is an operad in G spaces $\mathcal{O}(n)$ such that

- 1 Σ_n -acts freely,
- 2 if $\Gamma \subseteq G \times \Sigma_n$, the fixed points $\mathcal{O}(n)^\Gamma$ are either empty or contractible, and
- 3 the G -fixed points $\mathcal{O}(n)^G$ are always contractible.

Generalizing E_k ?

We don't even have a good definition for what an N_k operad should be!

Rewriting: Admissible Sets

Lemma (Graph Subgroups: ① & ②)

If $\Gamma \subset G \times \Sigma_n$ has $\Gamma \cap \Sigma_n = \{e\}$, then

$$\Gamma = \left\{ (h, f(h)) \mid h \in H \subset G, f: H \rightarrow \Sigma_n \right\}.$$

Definition

A finite H -set T is admissible for $\mathcal{O}$ if $\mathcal{O}(|T|)^{\Gamma_T} \neq \emptyset$ for Γ_T any graph subgroup classifying the H -set T .

Example (③)

The finite G -set $\underbrace{* \amalg \cdots \amalg *}_n$ is admissible for any n .

N_∞ operads do the right thing

Theorem (Blumberg–H.)

If T is an admissible H -set for $\mathcal{O}$, then $\mathcal{O}$ structures an essentially unique H -equivariant twisted multiplication

$$\mathrm{Map}(T, X) \rightarrow X.$$

Example

An ordinary $\mathcal{E}_\infty$ operad!

Only have a coherently commutative multiplication.

Example

Little disks for $(V \oplus \mathbb{R})^{\oplus \infty}$.

T is admissible iff $T \hookrightarrow V \oplus \mathbb{R}$.

Indexing compatibilities

Consequences

T being admissible for H implies the existence of lots of other admissible sets!

Example

The transfer $\text{Map}(C_2, \Omega X) \rightarrow \Omega^\sigma X$ gave the ordinary loop multiplication.

Theorem (Blumberg–H.)

If two N_∞ operads structure the same admissible sets, then they are equivalent up to homotopy.

Classification Theorem

Admissible Sets

Indexing systems form a poset $\mathcal{P}$.

Theorem (Blumberg–H.)

The assignment taking an N_∞ operad to its indexing system of admissible sets gives a fully faithful functor

$$ho\mathcal{N}_\infty \rightarrow \mathcal{P}.$$

Theorem (Bonventre–Pereira, Gutiérrez–White, Rubin)

Every indexing system is the indexing system associated to an N_∞ -operad.

Kervaire Invariant One

General Strategy

Build a cohomology theory that

- 1 detects Kervaire invariant one elements
- 2 has periodic homotopy groups
- 3 has lots of zeros in homotopy

Norms are in all parts

- 1 building the cohomology theory
- 2 showing the period and zeros

Twisted multiplications

Transfers and norms in algebra (Blumberg–H.)

- 1 Additive: incomplete Mackey functors
- 2 Multiplicative: incomplete Tambara functors
- 3 Both: Bi-incomplete Tambara functors

Equivariant Algebra!

Can carry out many of the classical algebra constructions and computations with these twisted rings!

Equivariant Algebra

Theorem (Schuchardt–Spitz–Wisdom)

The algebraically closed Tambara functors are in bijective correspondence with algebraically closed fields.

Theorem (Lindenstrauss–Richter–Zou)

Galois field extensions give families of étale extensions of Tambara functors.

Theorem (H.–Lawson–Wilson)

Étale-ness of a map of Tambara functors is insensitive to the norms.

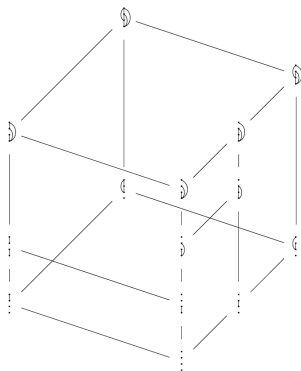
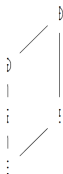
Transfer Systems

Definition (Rubin, Balchin–Barnes–Roitzheim)

A transfer system on G is a poset structure $\rightarrow$ on $\text{Sub}(G)$ satisfying

- 1 *If $K \rightarrow H$, then $K \subset H$,*
- 2 *if $K \rightarrow H$ and $g \in G$, then $gKg^{-1} \rightarrow gHg^{-1}$, and*
- 3 *if $K \rightarrow H$ and $J \subset H$, then $K \cap J \rightarrow J$.*

Bonus! Cyclic p -groups



Theorem (Balchin–Barnes–Roitzheim)

There are $\text{Cat}(n + 1)$ transfer systems for C_{p^n} .

Compatible Transfer Systems

Summary of structure

Transfer systems parameterize additive transfers and multiplicative norms.

Norms force transfers

(Twisted) multiplication distributes over addition, giving twisted additions:

$$\underbrace{(a + b) \cdot (\gamma a + \gamma b)}_{n_e^{C_2}(a+b)} = \underbrace{a \cdot \gamma a}_{n_e^{C_2}(a)} + \underbrace{b \cdot \gamma b}_{n_e^{C_2}(b)} + \underbrace{(a \cdot \gamma b + b \cdot \gamma a)}_{tr_e^{C_2}(a \cdot \gamma b)}.$$

Counting compatibles for C_{p^n}

Theorem (H.–Meng–Li–Ma)

There are

$$\frac{1}{3n+1} \binom{3n+1}{n}$$

compatible pairs of transfer systems for $C_{p^{n-1}}$.